Fiscal Year:	FY 2022	**Task Last Updated:**	FY 07/23/2025
PI Name:	Beheshti, Afshin Ph.D.		
Project Title:	miRNA Signature Detection and Countermeasures Against HZE Radiation Exposure for Tissue Degeneration		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	PI has moved from NASA Ames Research Center to the University of Pittsburgh, where he serves as Director of the Center for Space Biomedicine. Former email: afshin.beheshti@nasa.gov		
Project Type:	Ground	**Solicitation / Funding Source:**	2018 TRA BRASH1801: Translational Research Institute for Space Health (TRISH) Biomedical Research Advances for Space Health
Start Date:	01/01/2019	**End Date:**	08/31/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/31/2021 per TRISH Task Book submission (Ed., 7/2/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Blaber, Elizabeth Ph.D. (Rensselaer Polytechnic Institute) Costes, Sylvain Ph.D. (NASA Ames Research Center) Grabham, Peter Ph.D. (Columbia University)		
Grant/Contract No.:	NNX16AO69A-T0404		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project will investigate the hypothesis that circulating microRNA (miRNA) signatures are driving microvascular disease and muscle degeneration associated with accelerating aging and will be enhanced by exposure to the space environment both in vivo (mice) and in vitro and test novel antagonist therapies to these miRNA signatures as countermeasures to reduce space radiation-induced health risks. A comprehensive Systems Biology approach will be used. Galactic cosmic rays (GCR) and solar particle events (SPE) scenarios will be used. Lastly, antagonists to the miRNA signatures will be synthesized using novel self-delivery miRNA technology and administered to the murine model with space irradiation. This will allow the possible development of a novel minimally invasive miRNA based radioprotector to be used as a countermeasure for space radiation. Collectively, understanding of how whole body space radiation impacts microvascular and tissue degeneration through circulating miRNAs will greatly enhance health risk prognostication and provide possible new mechanisms for protection against space radiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	IMPACT: Collectively, the work being done will provide strong evidence for biologically measurable molecular signatures in delineating health risk caused by the space environment through a novel systems biology approach. The generated data will allow novel mechanistic understanding of how circulating miRNAs are the essential regulators that control health risk due to HZE (High-Ionizing High Energy Particles) irradiation as well as form the basis of International Space Station (ISS) experiments and eventually lead to potential novel radioprotectors that can be utilized for disease prevention in astronauts. INNOVATION: The current work is using an innovative systems biology approach to study how a predicted circulating miRNA signature will impact health risk due to space irradiation. We utilized simulated microgravity conditions during and after the space irradiation to fully simulate conditions during spaceflight. Overall, there is limited knowledge in the NASA community concerning the role of miRNAs on tissues and even less is known regarding circulating miRNAs impact on the host at a system level. Additionally, overall health risk due to the combined effects of microgravity and space radiation are not understood well. We are pursuing the biological impact of the circulating miRNA signature associated with microgravity and space irradiation conditions through cutting edge countermeasure techniques. Through the use of a self-deliverable antagomir to target the miRNAs associated with space radiation and cardiovascular risk we are starting to show a minimally invasive technique to mitigate space radiation damage. This work allows the creation of a possible countermeasure to protect against long-duration space related biological damage.
Task Progress:	• We have determined a miRNA signature that is associated with GCR radiation. • We have also determined a miRNA signature that is associated with SPE irradiation. • The above findings were done by miRNA-sequencing on mice with and without hindlimb unloading (HU) conditions (to simulated microgravity) irradiated with 0.5Gy simplified GCR Sim, 1Gy SPE sim, 5Gy gamma, and unirradiated control mice. The following tissues were sequenced: liver, heart, serum, and soleus muscle. • Countermeasure experiments with 3D microvasculature tissue model was performed. We show that, by inhibiting three of the miRNAs in this GCR signature, that we can mitigate the damage caused by GCR radiation. In vivo experiments have been conducted and show promising results with rescuing some impact on certain organs. • 5 publications related to Space Biology Research. • Distinct immune profile with space radiation identified and related to circulating miRNAs.
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